

Lab 3 - Routing Protocol

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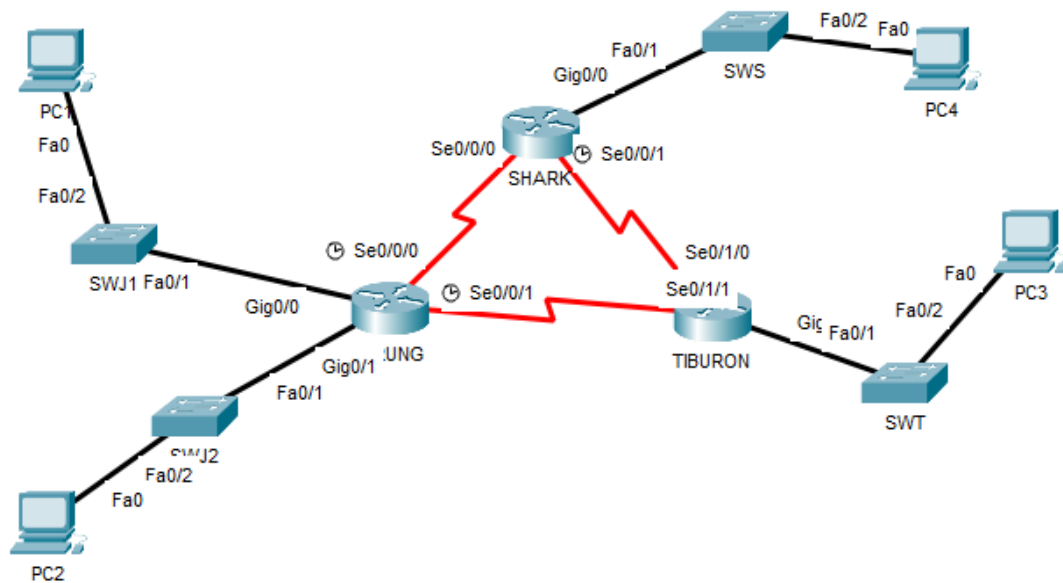
MATRIC NUM.: A23CS0265

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.129	255.255.255.224	-
4		G0/1	172.18.110.161	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.65	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.1	255.255.255.192	-
11	PC1	-	172.18.110.158	255.255.255.224	172.18.110.129
12	PC2	-	172.18.110.190	255.255.255.224	172.18.110.161
13	PC3	-	172.18.110.126	255.255.255.192	172.18.110.65
14	PC4	-	172.18.110.62	255.255.255.192	172.18.110.1

		128	64	32	16	8	4	2	1	
172.18.110.0/23										
Jerung LAN 1	$2^5 = 32 > 20+2 \therefore$ borrow bit = 5 bit									} $\therefore 172.18.110.0/27$
Jerung LAN 2	$2^5 = 32 > 30+2 \therefore$ borrow bit = 5 bit									
Shark LAN	$2^6 = 64 > 60+2 \therefore$ borrow bit = 6 bit									} $\therefore 172.18.110.0/26$
Tiburon LAN	$2^6 = 64 > 50+2 \therefore$ borrow bit = 6 bit									
Connection	$2^2 = 4 > 2+2 \therefore$ borrow bit = 2 bit									} $\therefore 172.18.110.0/30$
→ 172.18.01101110.00000000										
SHARK LAN	0.00 000000									NA : 172.18.110.0 BA : 172.18.110.63 usable host : 172.18.110.1 to 62
TIBURON LAN	0.01 000000									NA : 172.18.110.64 BA : 172.18.110.127 usable host : 172.18.110.65 to 126
JERUNG LAN 1	0.100 00000									NA : 172.18.110.128 BA : 172.18.110.159 usable host : 172.18.110.129 to 158
JERUNG LAN 2	0.101 00000									NA : 172.18.110.160 BA : 172.18.110.191 usable host : 172.18.110.161 to 190
JERUNG to SHARK	0.110000 00									NA : 172.18.110.192 BA : 172.18.110.195 usable host : 172.18.110.193 to 194
JERUNG to TIBURON	0.110001 00									NA : 172.18.110.196 BA : 172.18.110.199 usable host : 172.18.110.197 to 198
SHARK to TIBURON	0.110010 00									NA : 172.18.110.200 BA : 172.18.110.203 usable host : 172.18.110.201 to 202
J → S : SE 010/0 to SE 010/0 (193 to 194)										
J → T : SE 010/1 to SE 011/1 (197 to 198)										
S → T : SE 010/1 to SE 011/0 (201 to 202)										

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - a. CLI
 - i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

JERUNG

```
JERUNG>enable
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C    172.18.110.128/27 is directly connected, GigabitEthernet0/0
L    172.18.110.129/32 is directly connected, GigabitEthernet0/0
C    172.18.110.160/27 is directly connected, GigabitEthernet0/1
L    172.18.110.161/32 is directly connected, GigabitEthernet0/1
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.193/32 is directly connected, Serial0/0/0
C    172.18.110.196/30 is directly connected, Serial0/0/1
L    172.18.110.197/32 is directly connected, Serial0/0/1
```

SHARK

```
SHARK> enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.0/26 is directly connected, GigabitEthernet0/0
L    172.18.110.1/32 is directly connected, GigabitEthernet0/0
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.194/32 is directly connected, Serial0/0/0
C    172.18.110.200/30 is directly connected, Serial0/0/1
L    172.18.110.201/32 is directly connected, Serial0/0/1
```

TIBURON

```
TIBURON>enable
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.64/26 is directly connected, GigabitEthernet0/0
L    172.18.110.65/32 is directly connected, GigabitEthernet0/0
C    172.18.110.196/30 is directly connected, Serial0/1/1
L    172.18.110.198/32 is directly connected, Serial0/1/1
C    172.18.110.200/30 is directly connected, Serial0/1/0
L    172.18.110.202/32 is directly connected, Serial0/1/0
```

- b. PT tool
- Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

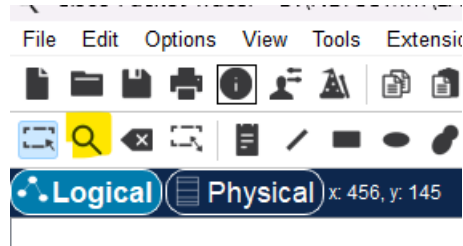


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

SUCCESS

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

3. Try to ping PC4 from PC1, paste the results here.

FAIL

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Request timed out.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

4. Explain the reason(s) behind the results.

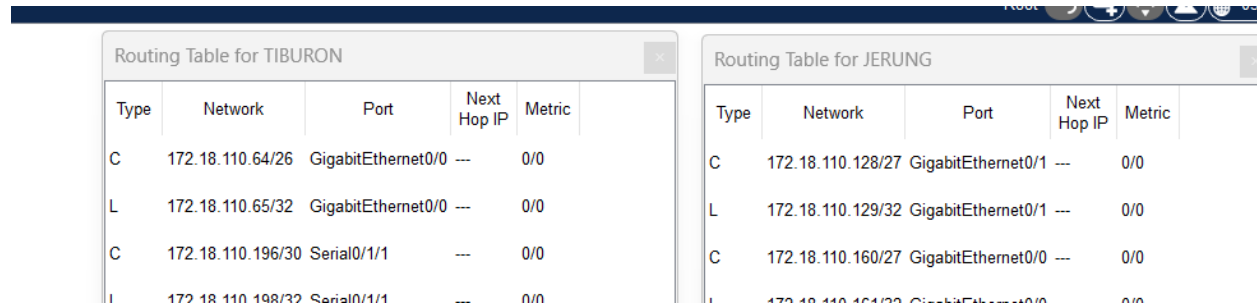
The PC 1 and PC 2 is connected successfully because the network connection between them is working correctly, and they are in the same subnet. However, PC4 and PC1 connected unsuccessfully as the network connection are not properly configured.

5. What needs to be done to ensure all PCs can ping each other successfully?

We need to configure the routers with proper routes to include all network segments. This will ensure that the router know how to forward traffic between the PCs.

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.



Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- What can you say about the addresses used in the 'network' instructions in Figure E?
The 'network' instructions is the network address (NA) of router JERUNG's interface. They also act as the router information of JERUNG.
- Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```

TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200

```

- As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

JERUNG

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 10 subnets, 4 masks
R    172.18.110.64/26 [120/1] via 172.18.110.198, 00:00:07, Serial0/0/1
C    172.18.110.128/27 is directly connected, GigabitEthernet0/0
L    172.18.110.129/32 is directly connected, GigabitEthernet0/0
C    172.18.110.160/27 is directly connected, GigabitEthernet0/1
L    172.18.110.161/32 is directly connected, GigabitEthernet0/1
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.193/32 is directly connected, Serial0/0/0
C    172.18.110.196/30 is directly connected, Serial0/0/1
L    172.18.110.197/32 is directly connected, Serial0/0/1
R    172.18.110.200/30 [120/1] via 172.18.110.198, 00:00:07, Serial0/0/1
```

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.64/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

TIBURON

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```

      172.18.0.0/16 is variably subnetted, 9 subnets, 4 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
R       172.18.110.128/27 [120/1] via 172.18.110.197, 00:00:25, Serial0/1/1
R       172.18.110.160/27 [120/1] via 172.18.110.197, 00:00:25, Serial0/1/1
R       172.18.110.192/30 [120/1] via 172.18.110.197, 00:00:25, Serial0/1/1
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

- a. What are the changes seen in TIBURON?

New routes generated by the RIP protocol are added into the routing table with a “R” sign that represents RIP.

- b. What are the Networks with type R in TIBURON and JERUNG?

These networks are managed by the Routing Information Protocol (RIP). It used to manage router information in a self- contained network such as a corporate LAN or private WAN. They will automatically be updated in the routers’ routing table.

- c. Ping PC3 from PC1. Was it successful?

Yes. Its success.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=26ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 26ms, Average = 9ms
```

- d. Ping PC4 from PC1. Was it successful?

No. It not success.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- e. Explain the reasons for your answer.

The ping from PC1 to PC3 was successful because the network connection between them is correctly configured and the routes are set up to allow communication. RIP routing protocol of JERUNG for PC1 and TIBURON for PC3 have properly setup and configured then automatically be updated in the routers’ routing table.

The ping from PC1 to PC4 was not successful because the network connection between them is not properly configured and the routing table of SHARK for PC4 has not properly setup and configured.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

- g. Open router SHARK's routing table and paste here.

```
enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 10 subnets, 4 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
R       172.18.110.64/26 [120/1] via 172.18.110.202, 00:00:19, Serial0/0/1
R       172.18.110.128/27 [120/1] via 172.18.110.193, 00:00:22, Serial0/0/0
R       172.18.110.160/27 [120/1] via 172.18.110.193, 00:00:22, Serial0/0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
R       172.18.110.196/30 [120/1] via 172.18.110.202, 00:00:19, Serial0/0/1
           [120/1] via 172.18.110.193, 00:00:22, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
--More--
```

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/1	
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	SUCCESSFUL <pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=7ms TTL=126 Reply from 172.18.110.158: bytes=32 time=2ms TTL=126 Reply from 172.18.110.158: bytes=32 time=11ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 11ms, Average = 5ms C:\> </pre>
2	SUCCESSFUL <pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=12ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=9ms TTL=126 Reply from 172.18.110.190: bytes=32 time=5ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 12ms, Average = 6ms </pre>

3	<pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=8ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=8ms TTL=126 Reply from 172.18.110.126: bytes=32 time=5ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 8ms, Average = 5ms </pre>
---	--

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0/24](#)
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?
[PC4 and PC1 cannot ping PC3 after this change.](#)

PC4 Fail

```

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.202: Destination host unreachable.
Reply from 172.18.110.202: Destination host unreachable.
Reply from 172.18.110.202: Destination host unreachable.
Reply from 172.18.110.202: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```


PC1 Fail

```

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.198: Destination host unreachable.
Reply from 172.18.110.198: Destination host unreachable.
Reply from 172.18.110.198: Destination host unreachable.
Request timed out.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

```

SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 9 subnets, 4 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
R       172.18.110.128/27 [120/1] via 172.18.110.193, 00:00:19, Serial0/0/0
R       172.18.110.160/27 [120/1] via 172.18.110.193, 00:00:19, Serial0/0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
R       172.18.110.196/30 [120/1] via 172.18.110.202, 00:00:22, Serial0/0/1
           [120/1] via 172.18.110.193, 00:00:19, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

```

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 * - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is not set

```

    172.18.0.0/16 is variably subnetted, 8 subnets, 4 masks
R       172.18.110.0/26 [120/1] via 172.18.110.201, 00:00:04, Serial0/1/0
R       172.18.110.128/27 [120/1] via 172.18.110.197, 00:00:06, Serial0/1/1
R       172.18.110.160/27 [120/1] via 172.18.110.197, 00:00:06, Serial0/1/1
R       172.18.110.192/30 [120/1] via 172.18.110.197, 00:00:06, Serial0/1/1
          [120/1] via 172.18.110.201, 00:00:04, Serial0/1/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
--More-- |

```

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1	
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1	
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1	
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1	
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	
L	172.18.110.202/32	Serial0/1/0	---	0/0	
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0	
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0	

- e. Referring to the routing table, explain your findings.

The new network address of the subnet has not updated into the routing table.

Thus, it causes all the packets that destined for PC3 to be dropped. As a result, PC4 and PC1 cannot ping PC3 after changing the new IP address.

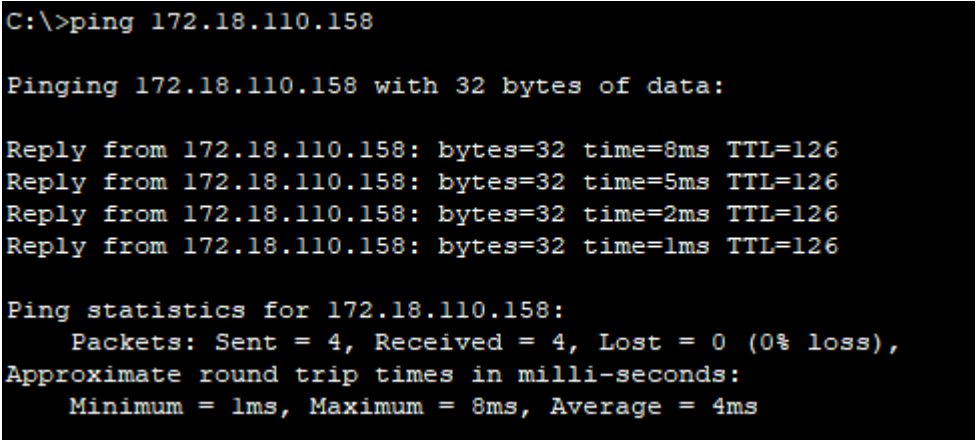
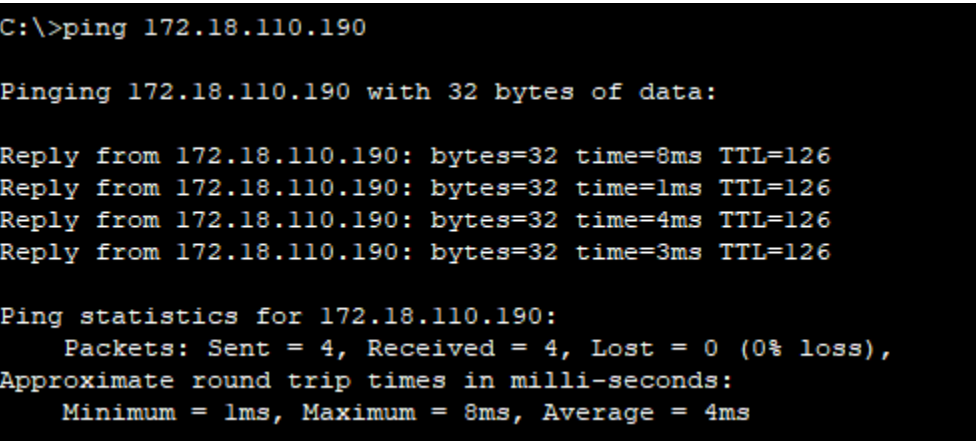
- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

I need to update the routing configuration in TIBURON to include the new subnet (192.168.1.0/24) and advertise it using the RIP protocol to ensure that it can be automatically updated into the routing table of TIBURON.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>
TIBURON>enable
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.196
TIBURON(config-router)#network 175.18.110.200
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	Successful  <pre>C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=8ms TTL=126 Reply from 172.18.110.158: bytes=32 time=5ms TTL=126 Reply from 172.18.110.158: bytes=32 time=2ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 8ms, Average = 4ms</pre>
2	Successful  <pre>C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=8ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=4ms TTL=126 Reply from 172.18.110.190: bytes=32 time=3ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 8ms, Average = 4ms</pre>

4	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=2ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=2ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms </pre>
---	---

REFLECTION

What have you learned in this task?

In this lab exercise, I have learned a lot about how networks work, especially when it comes to routing and subnetting. Setting up the routers using the RIP protocol showed me how important it is to configure them correctly so all the devices able to communicate with each other. I realized that even a small mistake, like missing a route in routing table can break connections between devices.

Troubleshooting ping failures between PCs helped me to understand how to figure out what's wrong in a network and fix it step by step. It was satisfying to see everything work after making the right changes!

Otherwise, I also learned how changing IP addresses can affect the network and how to update configurations to keep everything connected. This task showed me how networks are like puzzles as you need to get all the pieces in right place for everything to work smoothly.

Overall, it was a great hands-on experience that made the concepts much clearer and more practical.